

100502 CONTROL SYSTEMS

Module 1: introduction to control problem

- Industrial control example
- Mathematical models of physical systems
- Control hardware and their models
- Transfer function models of linear time – invariant systems
- Feedback control : open -loop and closed -loop systems
- Benefits of feedback
- Block diagram algebra
- Signal flow graph

Module 2: time response analysis

- Standard test signals
- Time response of first and second order systems for standard test inputs
- Application of initial and final value theorem
- Design specification for second - order systems based on the time – response
- Concept of stability
- Routh – Hurwitz criteria
- Relative stability analysis
- Root-locus technique
- Construction of root-loci

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Module 3: frequency -response analysis

- Relationship between time and frequency response
- Polar plots
- Bode plots
- Nyquist stability criterion
- Relative stability using Nyquist criterion
- Gain and phase margin
- Closed -loop frequency response

Module 4: introduction to controller design

- Stability steady accuracy
- Transient accuracy
- Disturbance rejection
- Insensitivity and robustness of control systems
- Root -loci method of feedback control design
- Design specifications in frequency – domain
- Frequency -domain method of design
- Application of proportional
- Integral and derivative controllers
- Lead and lag compensation in designs
- Analog and digital implementation of controllers

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Module 5: state variable analysis

- Concepts of state variables
- State space model
- Diagonalization of state matrix
- Solution of state equation
- Eigen values and stability analysis
- Concept of controllability and observability
- Pole -placement by state feedback
- Discrete -time systems
- Difference equation
- State -space models of linear discrete -time systems
- Stability of linear discrete -time systems